

(a) analytically, (b) single application of the trapezoidal rule, (c) composite trapezoidal rule with $n = 2$ and 4, (d) single application of Simpson's 1/3 rule, (e) composite Simpson's 1/3 rule with $n = 4$, (f) Simpson's 3/8 rule, and (g) composite Simpson's rule, with $n = 5$. For each of the numerical estimates (b) through (g), determine the true percent relative error based on (a).

19.4 Evaluate the following integral:

$$\int_{-2}^4 (1 - x - 4x^3 + 2x^5) dx$$

(a) analytically, (b) single application of the trapezoidal rule, (c) composite trapezoidal rule with $n = 2$ and 4, (d) single application of Simpson's 1/3 rule, (e) Simpson's 3/8 rule, and (f) Boole's rule. For each of the numerical estimates (b) through (f), determine the true percent relative error based on (a).

19.5 The function

$$f(x) = e^{-x}$$

can be used to generate the following table of unequally spaced data:

x	0	0.1	0.3	0.5	0.7	0.95	1.2
$f(x)$	1	0.9048	0.7408	0.6065	0.4966	0.3867	0.3012

Evaluate the integral from $a = 0$ to $b = 0.6$ using (a) analytical means, (b) the trapezoidal rule, and (c) a combination of the trapezoidal and Simpson's rules wherever possible to attain the highest accuracy. For (b) and (c), compute the true percent relative error.

19.6 Evaluate the double integral

$$\int_{-2}^2 \int_0^4 (x^2 - 3y^2 + xy^3) dx dy$$

(a) analytically, (b) using the composite trapezoidal rule with $n = 2$, and (c) using single applications of Simpson's 1/3 rule. For (b) and (c), compute the percent relative error.

19.7 Evaluate the triple integral

$$\int_{-4}^4 \int_0^6 \int_{-1}^3 (x^3 - 2yz) dx dy dz$$

(a) analytically, and (b) using single applications of Simpson's 1/3 rule. For (b), compute the true percent relative error.

19.8 Determine the distance traveled from the following velocity data:

t	1	2	3.25	4.5	6	7	8	8.5	9	10
v	5	6	5.5	7	8.5	8	6	7	7	5

(a) Use the trapezoidal rule. In addition, determine the average velocity.

(b) Fit the data with a cubic equation using polynomial regression. Integrate the cubic equation to determine the distance.

19.9 Water exerts pressure on the upstream face of a dam as shown in Fig. P19.9. The pressure can be characterized by

$$p(z) = \rho g(D - z)$$

where $p(z)$ = pressure in pascals (or N/m^2) exerted at an elevation z meters above the reservoir bottom; ρ = density of water, which for this problem is assumed to be a constant 10^3 kg/m^3 ; g = acceleration due to gravity (9.81 m/s^2); and D = elevation (in m) of the water surface above the reservoir bottom. According to Eq. (P19.9), pressure increases linearly with depth, as depicted in Fig. P19.9a. Omitting atmospheric pressure (because it works against both sides of the dam face and essentially cancels out), the total force f_t can be determined by multiplying pressure times the area of the dam face (as shown in Fig. P19.9b). Because both pressure and area vary with elevation, the total force is obtained by evaluating

$$f_t = \int_0^D \rho g w(z)(D - z) dz$$

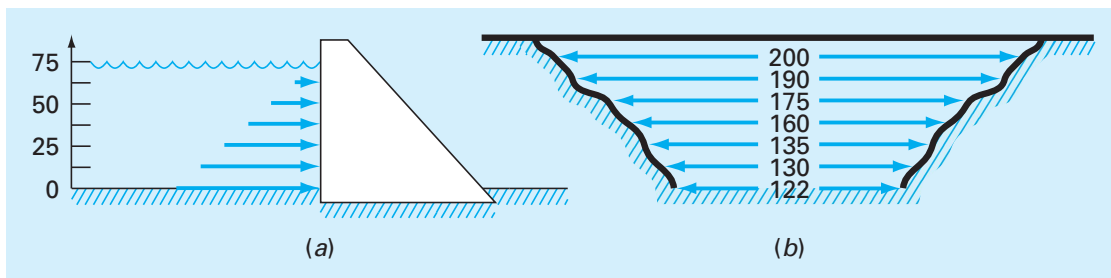


FIGURE P19.9

Water exerting pressure on the upstream face of a dam: (a) side view showing force increasing linearly with depth; (b) front view showing width of dam in meters.